

CSES Problem Set

Subordinates

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Given the structure of a company, your task is to calculate for each employee the number of their subordinates.

Input

The first input line has an integer n : the number of employees. The employees are numbered $1, 2, \dots, n$, and employee 1 is the general director of the company.

After this, there are $n - 1$ integers: for each employee $2, 3, \dots, n$ their direct boss in the company.

Output

Print n integers: for each employee $1, 2, \dots, n$ the number of their subordinates.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Example

Input:

```
5
1 1 2 3
```

Output:

```
4 1 1 0 0
```